Time-Frequency Analysis and Storage-Efficient Encoding of Real-Time Audio Signals Using STM32 Microcontroller

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***Abstract*—This paper presents an integrated framework for real-time acoustic signal acquisition, time–frequency analysis, and memory-efficient encoding on resource-constrained embedded platforms. An STM32 microcontroller equipped with a 12-bit analog-to-digital converter (ADC) is interfaced with a MAX4466 microphone module to acquire audio signals using hardware timer-triggered sampling. The acquired signals are characterized using Fast Fourier Transform (FFT) based spectrogram analysis to investigate both stationary and non-stationary acoustic characteristics. To address onboard storage constraints, two compression approaches are evaluated. The first employs 4-bit Adaptive Differential Pulse Code Modulation (ADPCM), achieving a 4:1 payload compression ratio with a signal-to-noise ratio (SNR) of 25.65 dB. The second combines Differential Pulse Code Modulation (DPCM) coding , ZigZag mapping, and zlib entropy coding delivering exact sample reconstruction (MSE = 0, SNR = infinity) at a 1.273:1 compression ratio. Unsupervised K-Means clustering (40-ms segments) and Support Vector Machine (SVM) classification with 97.08% accuracy validates segment-level signal properties. The complete architecture is targeted for industrial condition monitoring, automotive leak detection, and acoustic fault diagnosis.**

***Index Terms*—STM32 Microcontroller, MAX4466, Real-Time Audio Acquisition, Time–Frequency Analysis, STFT, Lossless DPCM, ADPCM Compression, Signal Reconstruction Quality.**

I. Introduction

Embedded audio acquisition requires accurate analog-to-digital conversion, deterministic sampling, and efficient memory management. STM32 microcontrollers provide integrated ADC peripherals with hardware triggering and DMA-based data transfer capabilities, enabling reliable acquisition of continuous sensor signals [7], [8]. The accuracy and configuration of ADC peripherals in STM32 devices have been discussed in STMicroelectronics application notes [9].

For embedded frequency-domain processing, optimized digital signal processing libraries such as CMSIS-DSP provide FFT, filtering, statistical, and machine-learning functions for ARM Cortex-M processors [10]. These capabilities support the implementation of computationally efficient signal-processing algorithms on resource-constrained microcontrollers.

Audio compression techniques such as Differential Pulse Code Modulation (DPCM) and Adaptive Differential Pulse Code Modulation (ADPCM) reduce the number of bits required to represent sampled signals [11], [12]. The IMA ADPCM recommendation provides a widely used 4-bit differential audio coding approach [13]. In addition, lossless compression can be achieved using predictive encoding and entropy coding techniques. The DEFLATE compression format combines LZ77-based dictionary matching and Huffman coding to reduce redundant data [14].Motivated by these techniques, this work integrates timer-triggered STM32 audio acquisition, FFT/STFT-based spectral characterization, lossy ADPCM compression, and lossless predictive encoding with ZigZag mapping and zlib compression within a unified embedded audio-processing framework.

II. System Architecture and Hardware Acquisition Pipeline

The STM32 ADC supports hardware-triggered conversion enabling efficient continuous signal acquisition with reduced CPU intervention [7], [8]. The ADC configuration and accuracy considerations are incorporated while designing the hardware , as discussed in ST Microelectronics application notes [7], [8].

A. Analog Front-End and Quantization Mechanics

The STM32 microcontroller employs a 12-bit ADC to digitize the analog audio signal obtained from the MAX4466 microphone amplifier module. The total number of discrete quantization levels is given by

$$L = 2^{12} = 4096. \tag{1}$$

For an ADC reference voltage of $V_{\mathrm{REF}} = 3.3$ V, the nominal voltage resolution per ADC code step is

$$\Delta V = \frac{V_{\mathrm{REF}}}{2^{12}}. \tag{2}$$

Substituting the reference voltage,

$$\begin{aligned} \Delta V &= \frac{3.3}{4096} \\ &\approx 0.8057 \ \mathrm{mV/code}. \end{aligned} \tag{3}$$

Assuming ideal uniform quantization, the quantization error e_q is uniformly distributed over the interval

$$-\frac{\Delta V}{2} \leq e_q \leq \frac{\Delta V}{2}. \tag{4}$$

The probability density function of the quantization error is expressed as

$$p(e_q) = \begin{cases} \dfrac{1}{\Delta V}, & -\frac{\Delta V}{2} \leq e_q \leq \frac{\Delta V}{2}, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

The quantization noise variance is calculated using

$$\sigma_q^2 = \frac{1}{\Delta V} \int_{-\Delta V/2}^{\Delta V/2} e_q^2 \, de_q. \tag{6}$$

Substituting the calculated ADC voltage resolution,

$$\begin{aligned} \sigma_q^2 &= \frac{(0.8057 \times 10^{-3})^2}{12} \\ &\approx 5.41 \times 10^{-8} \text{ V}^2. \end{aligned} \tag{7}$$

B. Quantization Noise Variance

For an ideal N-bit ADC with a maximum input voltage V_{MAX}, the voltage resolution is expressed as

$$\Delta V = V_{\text{MAX}} \times 2^{-N}. \tag{8}$$

Assuming uniformly distributed quantization error, the quantization noise variance is given by

$$\begin{aligned} \sigma_q^2 &= \frac{(\Delta V)^2}{12} \\ &= \frac{(V_{\text{MAX}} \times 2^{-N})^2}{12}. \end{aligned} \tag{9}$$

For a 12-bit ADC with a reference voltage of 3.3 V, the theoretical quantization noise variance becomes

$$\sigma_q^2 = \frac{(3.3 \times 2^{-12})^2}{12}. \tag{10}$$

This theoretical value provides a reference for assessing the quantization accuracy of the acquired audio signals.

C. Deterministic Hardware-Timer Triggering

To minimize sampling-time variations caused by software execution latency and interrupt scheduling, ADC conversions are directly triggered using a hardware timer depending on the selected STM32 device.The sampling frequency is determined by the effective timer clock frequency, prescaler (PSC), and auto-reload register (ARR), as expressed by

$$f_s = \frac{f_{\text{TIM}}}{(PSC+1)(ARR+1)}. \tag{11}$$

The timer period corresponding to the sampling frequency is given by

$$T_s = \frac{1}{f_s}. \tag{12}$$

For example, assuming an effective timer clock frequency of 84 MHz and a prescaler value of $PSC = 0$, the ARR values required for different sampling frequencies are calculated as follows.

For an 8-kHz sampling frequency,

$$ARR = \frac{84 \times 10^6}{8000} - 1 = 10499. \tag{13}$$

The hardware-timer-based acquisition mechanism provides uniform sampling intervals and reduces timing variations associated with software-controlled ADC triggering. This enables reliable time-domain and frequency-domain analysis of the acquired acoustic signals. The MAX4466 module provides the analog audio signal, while the STM32 microcontroller performs acquisition and digital processing.

III. Hardware Design and Implementation

The principal hardware components include the MAX4466 microphone amplifier module, STM32 development board, ADC peripheral, hardware timer.

A. ADC Resolution

For an ideal 12-bit ADC, the total number of quantization levels is given by

$$L = 2^{12} = 4096. \tag{14}$$

The approximate voltage increment per ADC code is

$$\Delta V = \frac{V_{\text{REF}}}{4096}, \tag{15}$$

where V_{REF} represents the ADC reference voltage.

B. Timer-Based Sampling

The timer-trigger frequency can be expressed as

$$f_s = \frac{f_{\text{TIM}}}{(PSC+1)(ARR+1)}, \tag{16}$$

where f_{TIM} is the effective timer clock, PSC is the prescaler register value, and ARR is the auto-reload register value.The timer clock must be determined from the clock configuration of the selected STM32 device.

IV. Spectral Analysis

FFT-based processing is computationally efficient compared with direct DFT computation [1], [2]. For embedded ARM Cortex-M implementations, optimized DSP functions are available through the CMSIS-DSP library [9], [10].

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}. \tag{17}$$

The Fast Fourier Transform (FFT) provides an efficient computational implementation of the DFT.

The frequency resolution is determined by

$$\Delta f = \frac{f_s}{N}, \tag{18}$$

where f_s is the sampling frequency and N is the transform length. For example, considering a sampling frequency of 16 kHz and an FFT length of 1024 samples,

$$\Delta f = \frac{16000}{1024} = 15.625 \text{ Hz}. \tag{19}$$

The resulting frequency spectrum can be used to identify dominant frequency components, fundamental frequencies, and associated harmonics.

V. Memory-Optimized Audio Compression

Differential Pulse Code Modulation (DPCM) exploits the correlation between consecutive audio samples to reduce the number of bits required for signal representation [11]. Adaptive Differential Pulse Code Modulation (ADPCM) further improves coding efficiency through adaptive quantization [12], [13].

A. Pulse Code Modulation

In Pulse Code Modulation (PCM), analog audio signals are sampled and quantized into digital values. For a 12-bit ADC, each sample is represented using 12 bits, although practical memory storage may require 16 bits per sample.

B. Differential Pulse Code Modulation

Differential encoding represents the difference between consecutive samples and is expressed as

$$d[n] = x[n] - x[n-1]. \tag{20}$$

The resulting differential samples may have a smaller dynamic range than the original samples. These values can subsequently be quantized and encoded using appropriate variable-length or entropy-coding techniques. Differential encoding alone does not guarantee a reduction in storage requirements. The actual compression performance depends on the quantization and encoding methods employed.

C. Compression Ratio

The compression ratio is defined as

$$CR = \frac{S_{\text{raw}}}{S_{\text{compressed}}}, \tag{21}$$

where S_{raw} and $S_{\text{compressed}}$ represent the original and compressed data sizes, respectively. A compression ratio of 4:1 is considered as the target for reducing storage requirements while maintaining acceptable audio signal characteristics.

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VI. Experimental Results and Discussion

The performance of the proposed STM32-based audio processing framework is evaluated using signal acquisition, frequency-domain analysis, time–frequency characterization, compression efficiency, and classification accuracy. The experimental evaluation considers sampling frequency, FFT execution time, memory utilization, processing latency, and reconstruction quality.

A. Audio Signal Acquisition

The audio signal acquired through the MAX4466 microphone module is digitized using the STM32 ADC as shown in Fig. 1. The Fig. 2 illustrates the acquired time-domain audio waveform. The waveform provides an overview of the amplitude variations and temporal characteristics of the input acoustic signal.

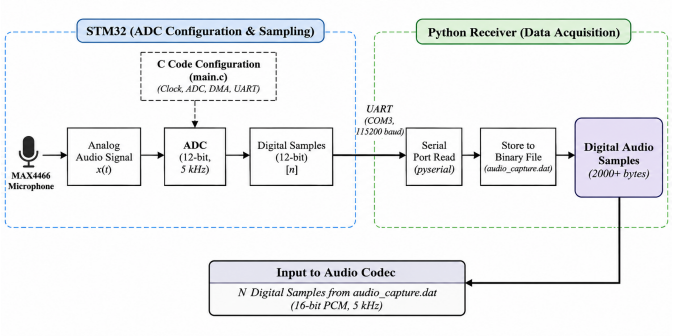


Fig. 1. Audio Acquisition and Compression setup.

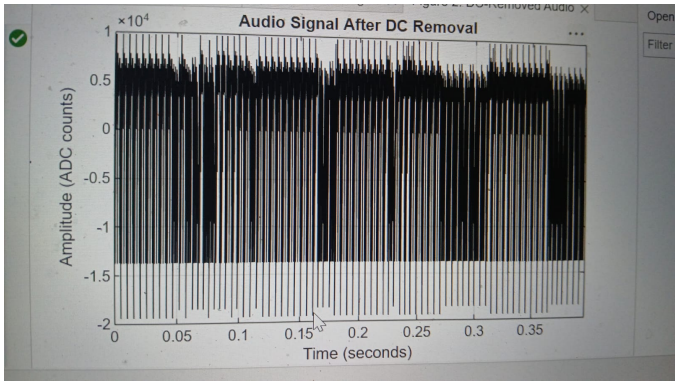


Fig. 2. Time-domain waveform of the acquired audio signal.

B. Frequency-Domain Analysis

The acquired audio samples are processed using the Fast Fourier Transform (FFT) to identify dominant frequency components and associated harmonics. Fig. 3 presents the frequency spectrum obtained from the acquired audio signal.

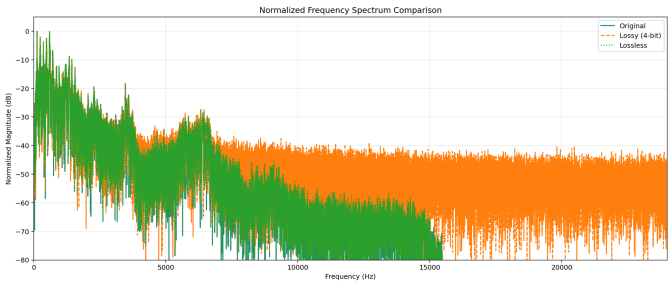


Fig. 3. FFT magnitude spectrum of the acquired audio signal.

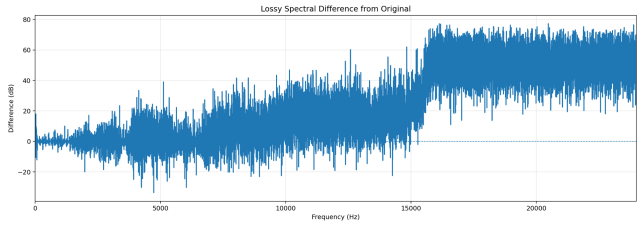


Fig. 4. Comparison of the original and reconstructed audio signals using ADPCM compression.

C. Audio Compression and Reconstruction

The compression performance is evaluated using 4-bit Adaptive Differential Pulse Code Modulation (ADPCM) and a lossless encoding approach based on Differential Pulse Code Modulation (DPCM), ZigZag mapping, and zlib compression.

Fig. 4 compares the original audio waveform with the reconstructed waveform obtained using the ADPCM compression technique. The reconstruction quality is evaluated using mean squared error (MSE) and signal-to-noise ratio (SNR).

The lossless compression pathway is evaluated by comparing the original and reconstructed sample sequences. Fig. 5 illustrates the reconstruction accuracy of the proposed lossless encoding approach.

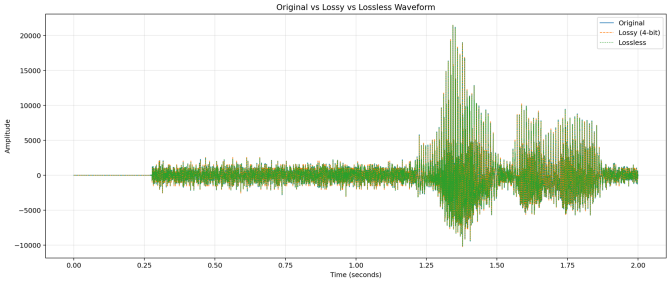


Fig. 5. Original and reconstructed signals using lossless encoding.

D. Classification Performance

K-Means clustering and Support Vector Machine (SVM) classification are investigated for segment-level acoustic characterization. Fig. 6 illustrates the classification results obtained from the evaluated acoustic dataset.

E. Performance Evaluation

VII. Conclusion

This paper presents a microcontroller-based framework for real-time audio acquisition, spectral analysis, and storage-efficient encoding using an STM32 platform. The proposed approach utilizes a MAX4466 microphone module, timer-triggered ADC conversion, and DMA-based data acquisition to obtain audio samples at consistent sampling intervals.

FFT-based spectral analysis enables the identification of dominant frequency components and harmonics, which can be useful for acoustic characterization and fault detection applications. PCM and differential encoding techniques are considered for reducing memory requirements. The target compression ratio is 4:1; however, its achievement must be verified through experimental implementation and signal-quality evaluation. Future work will focus on implementing the complete system on hardware, measuring processing performance, evaluating compression efficiency, and validating its suitability for real-time industrial acoustic monitoring.

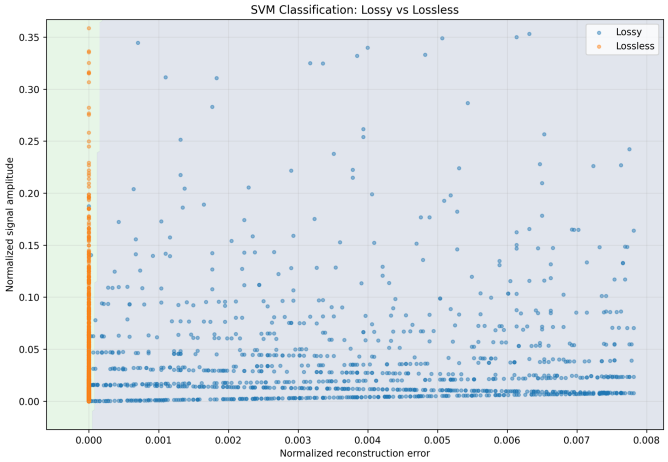


Fig. 6. SVM-based classification results for acoustic signal segments.

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